

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

COMPUTER GRAPHICS

Presented by: Eng.Karama

On Facebook, Twitter, Instagram:

@acmelacademy or
email:acmelacademy@gmail.com

SCOPE

FILLED AREA PRIMITIVES

Inside Test Overview

Odd-Even Rule

Filled Area Primitives Overview

Filled Area Algorithms

Scan line Fill Algorithm Explained .



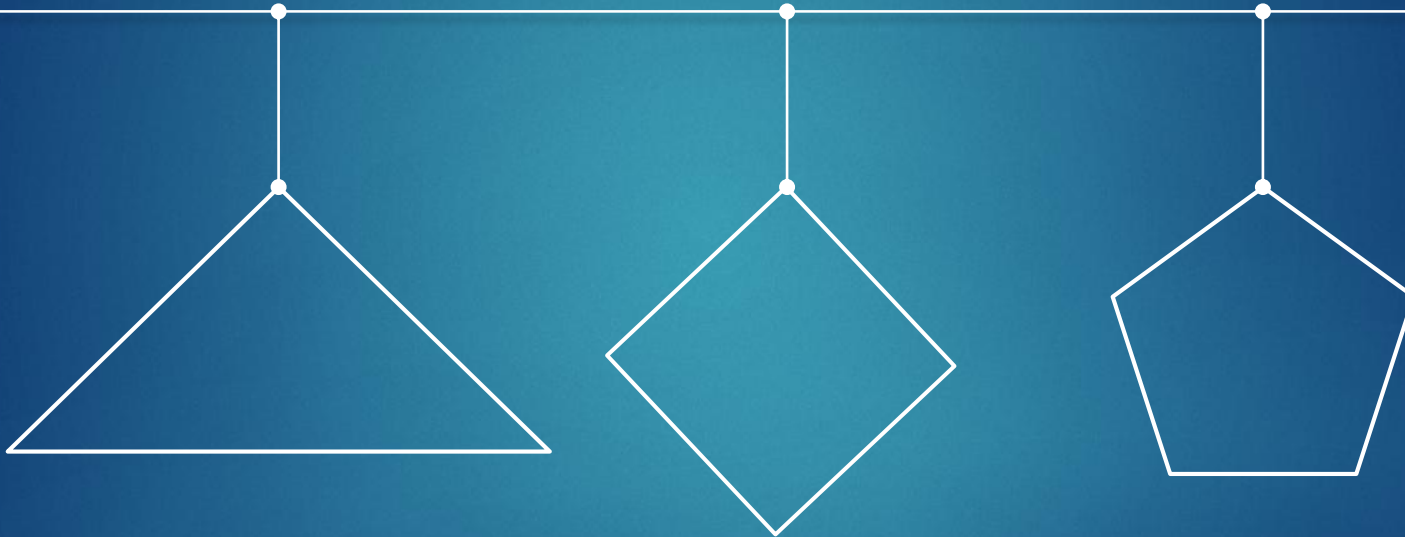
FILLED AREA PRIMITIVES EXPLAINED



© 2013 - 2026 akmel online

INSIDE TEST OVERVIEW?

Is to determine Which point(pixel) is inside the polygon or outside the polygon to make a polygon filling by using off-even rule.



ODD-EVEN RULE?

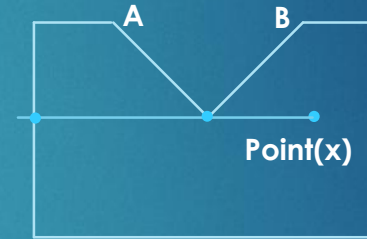
To determine whether a point(x) is inside or not. Draw a line starting from **Point(x)** to a **distant position**. **Count** the number of **edges** that **crosses** or **intersects** this line . If the count is **odd** then the point is **inside**, otherwise it is **outside**.



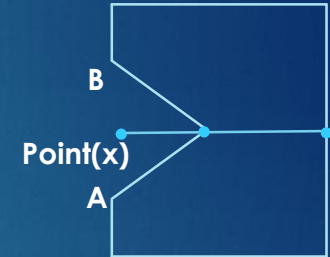
The count of intersection points are 3(odd)
Therefore this point is inside



The count of intersection points are 2(even)
Therefore this point is outside



The count of intersection points are 3(odd) and it is inside.
Because of, if the joining lines are same side of the point(x) line then read twice else read ones therefore the lines A and B are same side of line point(x).

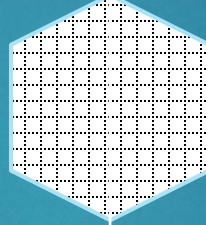


The count of intersection points are 2(even) and it is outside.
Because of, if the joining lines are different side of the point(x) line then read once else read twice therefore the lines A and B are different side of line point(x).



FILLED AREA PRIMITIVES OVERVIEW

is to fill the polygons with a specific color.



➤ filled area algorithms

- Scan line fill algorithms

- Seed fill algorithms

- - Boundary fill
- - Flood fill

- ❖ To fill the polygon first you need to:
 1. Find all interior pixels of the polygon
 2. Fill pixels with specified color



SCAN LINE FILL ALGORITHM

Is used to fill the polygon by intersecting scan line with polygon edges and fills the polygon between pairs of intersection.

BASICS

Location the intersection of the scan line with edges

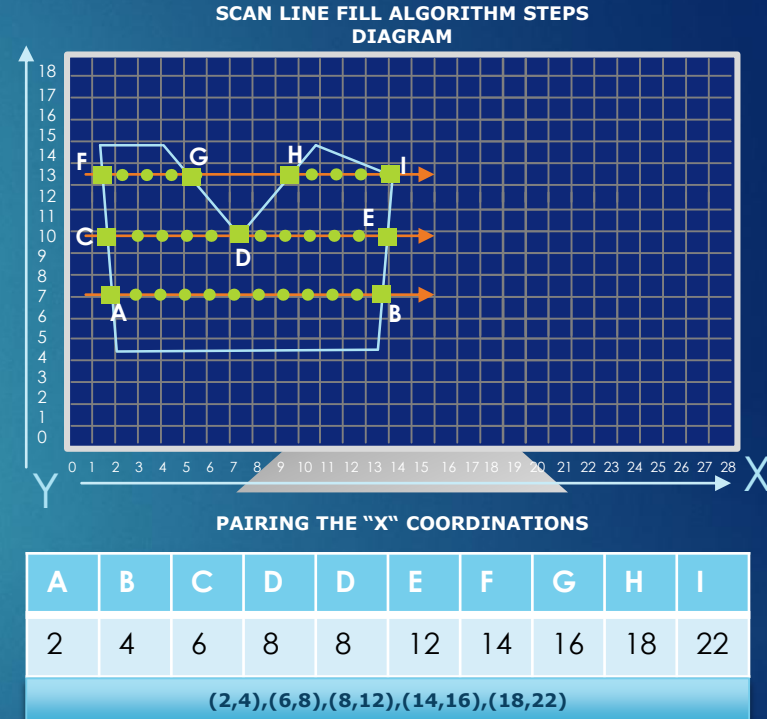
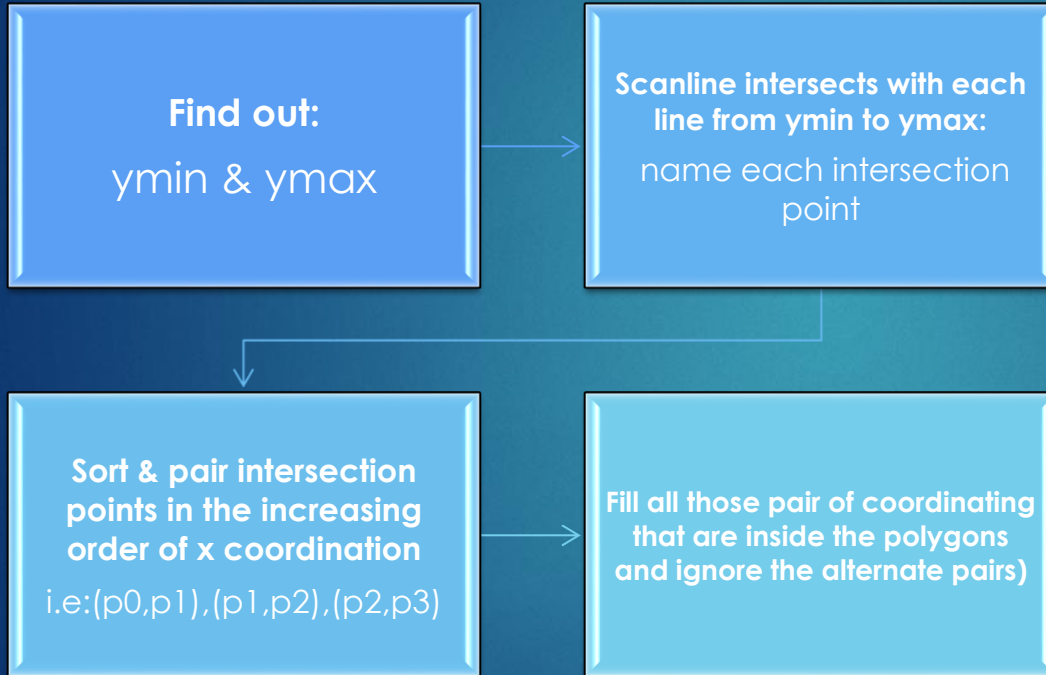
Sort the intersection points from left to right

Draw the interior intersection points pairwise

If the edges of the polygon lie on a different sides, count it once else count it twice



SCAN LINE FILL ALGORITHM STEPS





**MORE PRACTICE MAKES
YOU PERFECT**





Always Seek Knowledge



THE END

